

Evidence Brief: Data-Driven Content Performance Prediction

External proof points from media + tech

- 1) Peer-reviewed study (Computers & Electrical Engineering, 2022):
One-Class SVM + Autoencoder on UCI online news dataset achieved ~88% accuracy for popularity prediction.
- 2) Academic study (Journal of Media Horizons, 2025):
XGBoost achieved F1=0.697 and 69.8% accuracy on ~39k articles;
multimedia and weekend publication associated with higher popularity.
- 3) Industry case study (Taboola Newsroom):
Headline A/B tests increased CTR by >44% on average; several tests >200%.
- 4) Industry case study (Refinery29 + Taboola Newsroom):
Homepage CTR increased 45% per story on average; gains up to 200%.
- 5) Streaming tech (Netflix; Harvard D3):
Netflix estimated ~75% of member viewing comes from recommendations.

Implication for ASU:

Evidence shows measurable lifts from predictive modeling + content optimization in adjacent industries, supporting a pilot for ASU content planning.

Sources:

- ScienceDirect (2022): <https://www.sciencedirect.com/science/article/abs/pii/S0045790622004396>
- NYCU Academic Hub (2022): <https://scholar.nycu.edu.tw/en/publications/predicting-online-news-popularity-ba>
- Journal of Media Horizons (2025): <https://jmhorizons.com/index.php/journal/article/view/265>
- Taboola Newsroom case (CTR +44% / 200%+): <https://www.taboola.com/marketing-hub/a-b-headline-tests-bo>
- Taboola Refinery29 case (CTR +45% / 200%): <https://www.taboola.com/resources/case-studies/refinery29>
- Harvard D3 Netflix case (75% from recs): <https://d3.harvard.edu/platform-digit/submission/netflix-bringing-data>